

Spatial Correlation Between Quartz C-Axis Orientation and Strain Distribution in Metamorphic Quartzites

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Abstract

Micro-scale GIS mapping alongside structural geology data was projected in order to visualize strain in metamorphic quartzite thin sections. Multiple pictures of a quartzite thin-section were taken and had the orientation of the longest axis in the crystal (the c-axis) mapped to see trends in their angles, which would reveal microscopic shear zones. Many of these shear zones ran east to west, paralleling the general strain visible upon observation of the thin section. In further investigation on this topic, trace element presence in this thin section will be evaluated with S.E.M. analysis, and also incorporated into the GIS map of the quartzite.

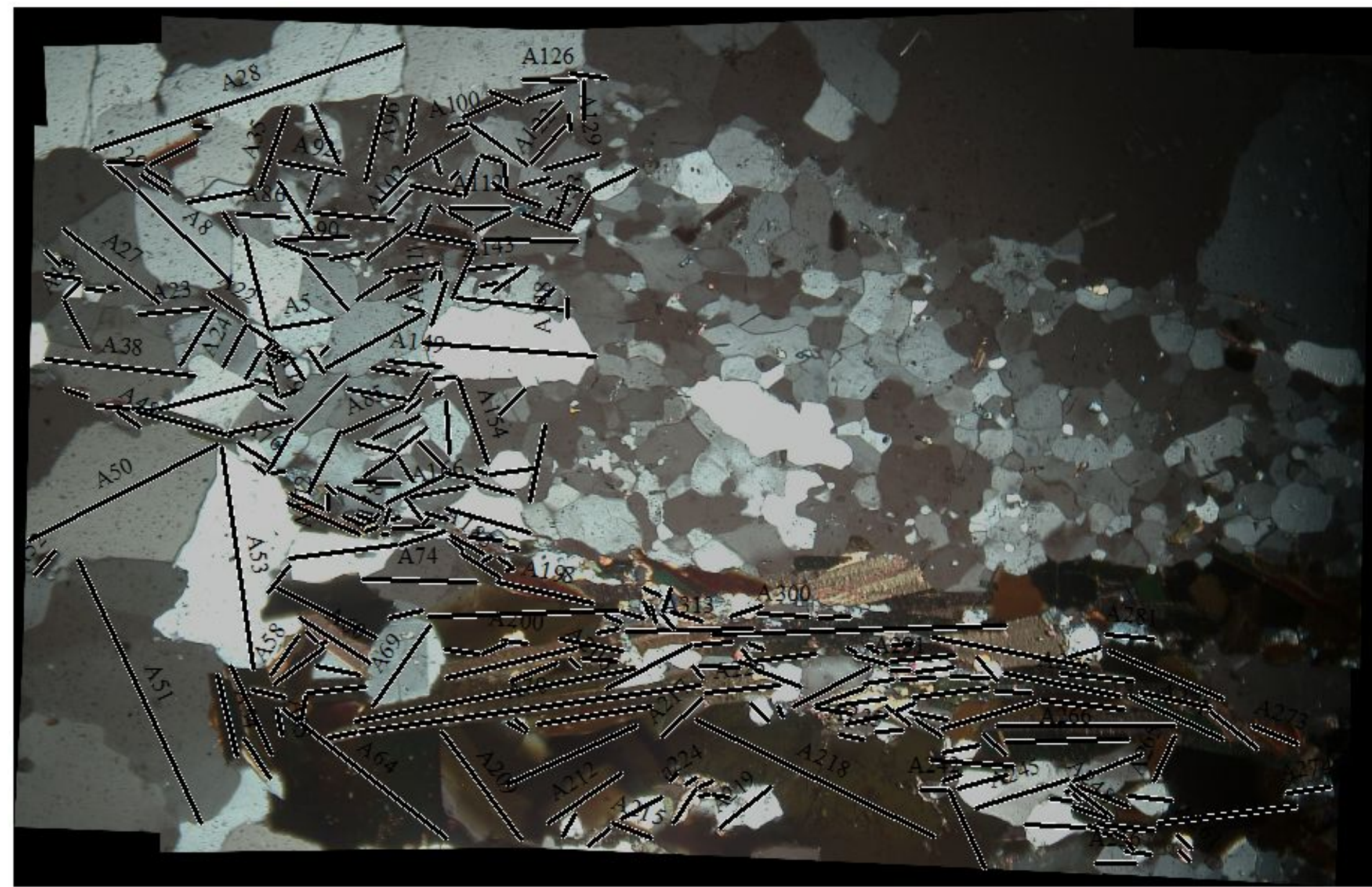


Figure 1a: How the c-axis mapping looks on the thin section. The white, grey, and black grains are quartz crystals. Other crystals are present as minor minerals or trace elements, like muscovite. Magnification is under x4 objective lens.

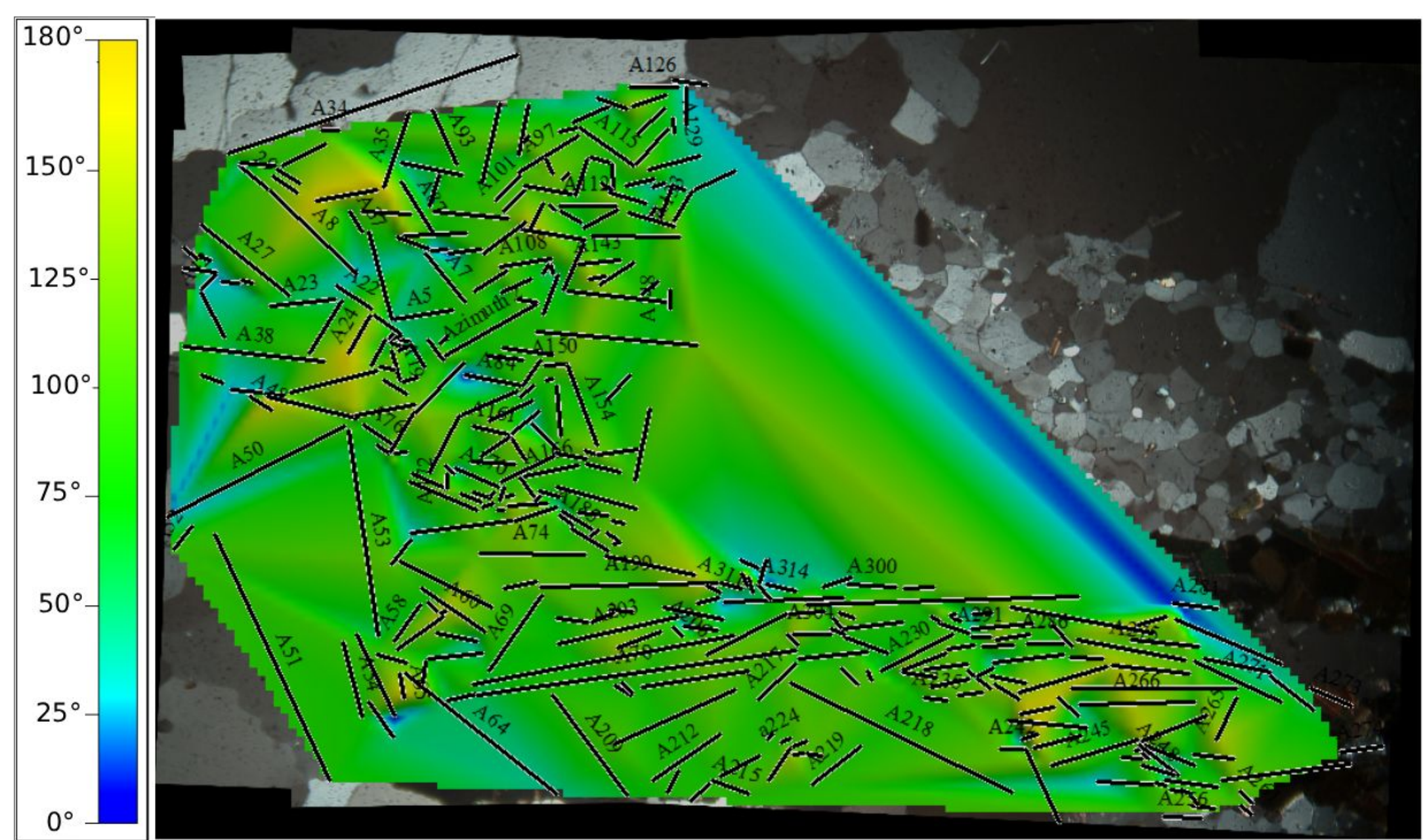


Figure 1b: How the c-axis mapping looks after contour conversion. The green areas symbolize a lot of horizontal strain, and small outlier areas yellow/blue colors appear to be different shear zones.

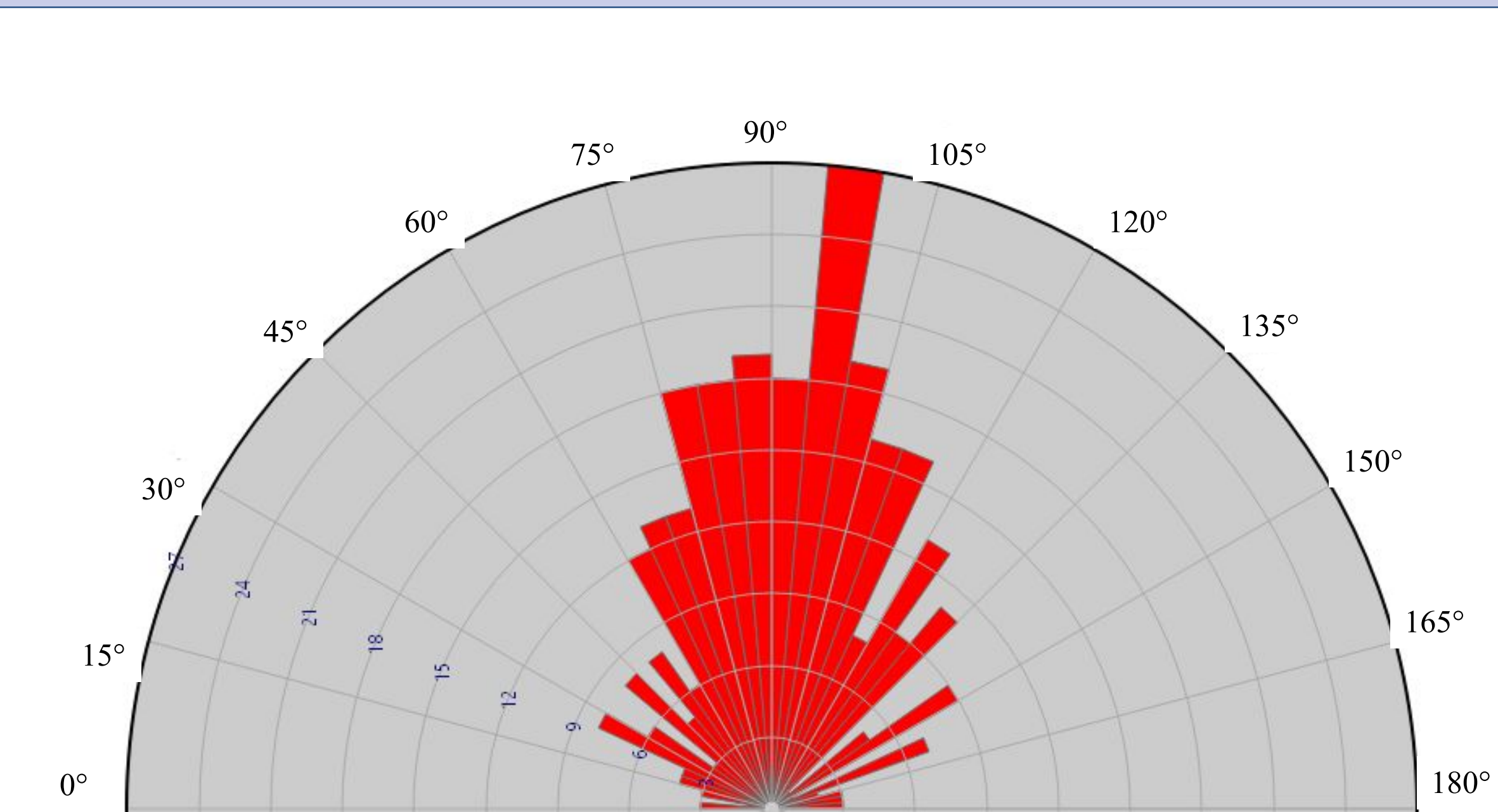


Figure 2: A rose diagram displaying the frequency of azimuth/c-axis throughout the sample. The most common lineation is between 75 and 105 degrees, making a sideways-strained quartz crystal most common in the sample.

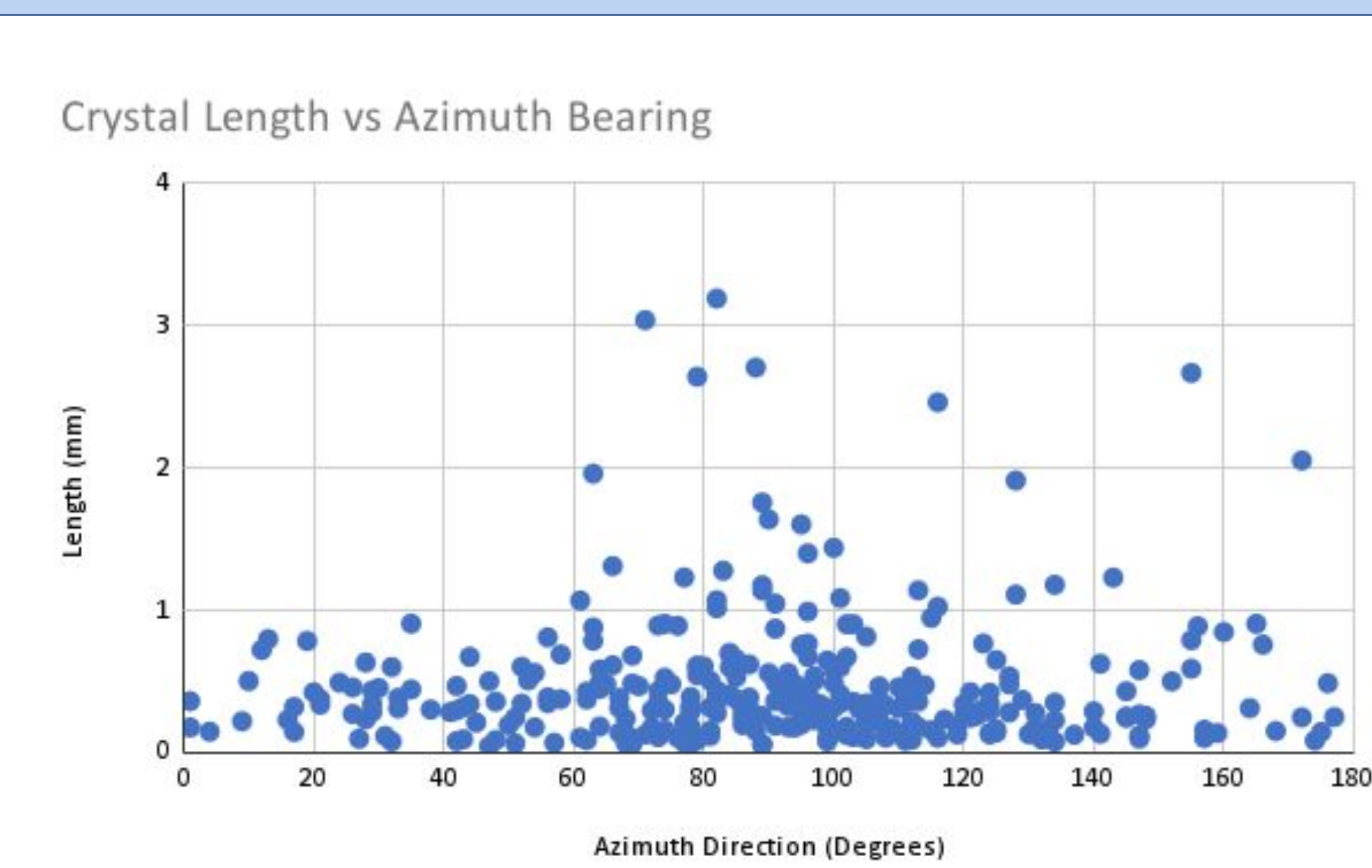


Figure 3: How crystal length correlates with a certain azimuth direction. Of note is that the longest crystals correspond to the established preferred strain direction visible in Figures 1b and 2.

Introduction

Metamorphic rocks are exposed to lots of heat and pressure to form, and strain on these rocks can be manifested as shear, which the visible deformation of rock caused by forces of heat/pressure acting upon it. These shear zones are also present on the microscopic level and are indicative of very localized metamorphic events that are only visible on a small portion of rock; but might be able to indicate that trace elements tend to aggregate in certain patterns inside rock when these conditions are present. These trace elements are a valuable economic resource and also provide ways to infer the history of the rock as well, making their presence important and worth analyzing. Trace elements get incorporated inside individual crystals as they cool, though the time this can happen is limited and they may stay isolated or incorporated into a defect in the crystal (Buseck & Veblen, 1978). Quartzite was chosen as it has a very uniform major element composition compared to other metamorphic rocks, allowing for trace elements to stand out much more. Microscopic quartz crystals also have a signature pattern, they will deform perpendicular to any stress they experience, in a similar manner to squeezing a sandwich with both your hands (Kamb, 1959). The longest axis of a deformed crystal that undergoes this stress is known as the c-axis.

Methodology

- The quartzite thin section was observed with a petrographic microscope and photographed with a microscope camera. Multiple overlapping pictures were taken of the section.
- Once on a computer, an image editor was used to stitch all these images together to form a big panorama.
- GlobalMapper was used (GIS) to get the direction of the c-axis in azimuths, drawing a line along each crystal. After this, each line was converted into a point. Each azimuth value was converted into an elevation value associated with a point.
- With all the azimuths equated to elevation, a contour map was made so all crystals with similar c-axis will be colored the same, allowing for shear zones to be made apparent.
- Also to be made is a rose diagram graphing out the distribution of c-axis among crystals, using GeoRose.

Ongoing Results/Discussion

- The azimuth map (Figure 1b) and rose diagram (Figure 2) show that the quartzite microscopic strain mainly runs horizontally, with some zones in the section having sharper angles.
- Figure 3 shows that the longest crystals are aligned in the most prevalent direction of strain.
- There are a few crystals in the sample that are relatively circular, suggesting that they did not deform as much as nearby crystals.
- Little pockets of crystals in this sample deform in different directions, suggesting that shear acted in multiple ways on this section of quartzite.

Conclusions / Further Steps

- The current findings show that using GIS to map thin-sections can yield interesting interpretations of microscopic shear zones.
- When mapping of the section is completed, a larger area of the sample will be able to show shear zones that could display an important trend.
- Incorporating S.E.M. analysis into this section can yield important information about how trace elements distribute in the microscopic setting. Correlating it with shear can also show how trace elements settle when placed under metamorphic conditions. These trends could then be applied to a larger-scale to give more methods for locating these valuable trace elements.
- Another interesting step to take is apply chemical mapping to this G.I.S. map to further understand how trace elements distribute among strained crystals.

References

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2. Kamb, W. B. (1959). Theory of Preferred Crystal Orientation Developed by Crystallization under Stress. *The Journal of Geology*, 67(2), 153–170. <https://doi.org/10.1086/626571>